

Chapter-3

Orthogonality

orthogonal vectors, orthogonal subspaces

1) length of a vector (mag)

let $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$

$$\|x\| = \text{"norm" of } x = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

$$\|x\| \geq 0$$

2) Inner product of two vectors (dot pro, scalar pro)

let $x, y \in \mathbb{R}^n$

$$x = (x_1, \dots, x_n), \quad y = (y_1, \dots, y_n)$$

$$\langle x, y \rangle = (x, y) = x \cdot y = x^T y = y^T x$$

$$= (x_1, x_2, \dots, x_n) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n = \text{scalar}$$

$$x^T x = x_1^2 + x_2^2 + \dots + x_n^2 = \|x\|^2$$

3) Orthogonal vectors

If $x^T y = 0$, then $x \perp y$.

If $x \perp y$, then $x^T y = 0$

for $v_1, v_2, \dots, v_k$

$$\boxed{v_i^T v_j = 0} \text{ for } i \neq j$$

$$\begin{array}{ll} \text{If } x^T y > 0, & < 90^\circ \\ x^T y < 0, & > 90^\circ \end{array}$$

Problem set 3-1

19. Which pairs are orthogonal among the vectors v_1, v_2, v_3, v_4 ?

$$v_1 = \begin{pmatrix} 1 \\ 2 \\ -2 \\ 1 \end{pmatrix} \quad v_2 = \begin{pmatrix} 4 \\ 0 \\ 4 \\ 0 \end{pmatrix} \quad v_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \quad v_4 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$(1 \ 2 \ -2 \ 1) \begin{pmatrix} 4 \\ 0 \\ 4 \\ 0 \end{pmatrix} = (4 + 0 - 8 + 0) \neq 0 \quad v_1 \not\perp v_2$$

$$(1 \ 2 \ -2 \ 1) \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} = 1 - 2 + 2 - 1 = 0 \quad v_1 \perp v_3$$

$$(1 \ 2 \ -2 \ 1) \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = 1 + 2 - 2 + 1 \neq 0 \quad v_1 \not\perp v_4$$

$$v_2 \perp v_3, \quad v_2 \not\perp v_4, \quad v_3 \not\perp v_4$$

Not orthogonal set.

If a set of vectors are orthogonal, then they must be linearly independent but the converse need not be true.

Orthogonality $\Rightarrow$ Independence

Independence $\nRightarrow$ orthogonality

$$^{\mathbb{R}^2} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\} \quad 0, 1$$

dependent $\rightarrow$ cannot be orthogonal.

$$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\} \quad \text{as } 1, 0$$

4) Unit vector :- Magnitude is 1.

$$\hat{v} = \frac{v}{\|v\|}$$

normalising a vector.

$$v = (1, 2, 1)$$

$$\hat{v} = \left(\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right)$$

5) Orthogonal ^{normal} ~~and~~ vectors :-

orthogonal unit vectors

$$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\} \rightarrow \left\{ \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}, \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \right\}$$

orthogonal basis

orthonormal basis

$$\rightarrow \begin{matrix} \text{Standard Basis} \\ \text{Orthonormal Basis} \end{matrix} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

Q. How to convert any basis into orthonormal basis?

* Orthogonal subspaces $\therefore$ not unique.

Let $\mathbb{R}^n$ is a vector space.

Let V and W be two subspaces of $\mathbb{R}^n$.

$$V \perp W = \{ v^T w = 0 \text{ for every } v \in V, w \in W \}$$

$\mathbb{R}^3$, x, y, z axis - trivial orthogonal subspaces.

Q. $\mathbb{R}^3$, ~~can~~ can we find two planes orthogonal in $\mathbb{R}^3$.

In $\mathbb{R}^3$,

Two planes in $\mathbb{R}^3$ cannot be orthogonal.

$\mathbb{R}^n$

row space $\perp$ nullspace
 r $n-r$

$\mathbb{R}^m$

Col space $\perp$ Left Null
 r $m-r$

* Orthogonal complement $\therefore$ unique

Subspace V of $\mathbb{R}^n$, the space of all vectors orth to V . ($V^\perp$)

$V \in \mathbb{R}^n$, $V^\perp$ such that

$$V^\perp \cup V = \mathbb{R}^n$$

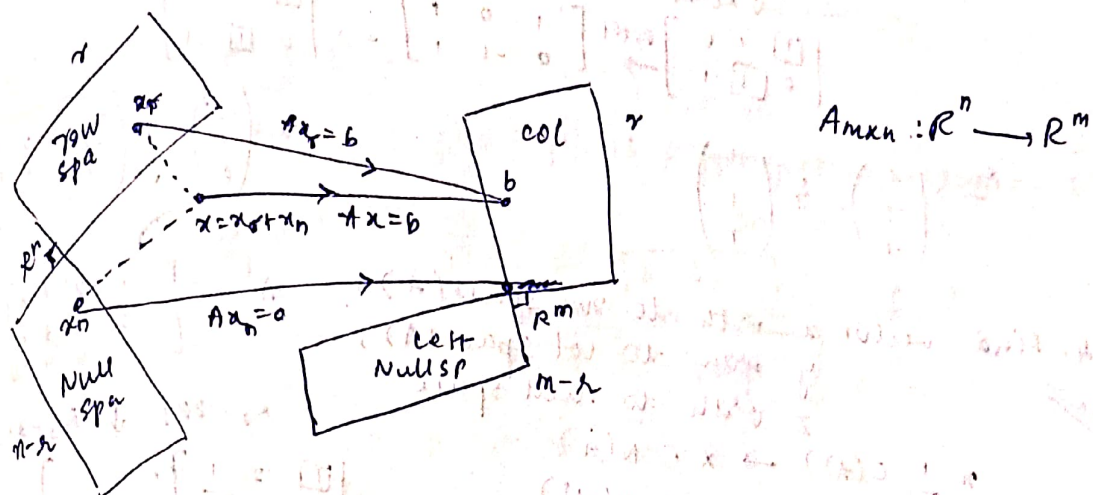
$$V^\perp \cap V = \{0\}$$

col space is left null space.

Fundamental theorem of LA

Null space is ortho comp of row space in $\mathbb{R}^n$ and
col space is ortho comp of left null space in $\mathbb{R}^m$.

Note: If S and T are ortho comp in $\mathbb{R}^n$ then
 $\dim(S) + \dim(T) = n \rightarrow$ always true.



To find parts of vectors in particular plane $\rightarrow$ project vector onto a line subspace.

Matrix and subspace :-

Project 'x' onto null space to find x_n , etc.

$Ax=b$ (Solvable) $\xrightarrow{\text{only if}} y^T b = 0$ whenever $y^T A = 0$

col space vectors $\rightarrow$ come from one vector from row space.

when matrix is invertible.

into : codomain has add

onto : codomain is range

Problem Set 3.1

1. Find length & inner product of $x = (1, 4, 0, 2)$, $y = (2, -2, 1, 3)$

a) $\|x\| = \text{norm of } x = \sqrt{1^2 + 16 + 0 + 4} = \sqrt{21}$

$\|y\| = \sqrt{4 + 4 + 1 + 9} = \sqrt{18} = \sqrt{2 \times 9} = 3\sqrt{2}$

b) Inner prod = $\langle x, y \rangle = x \cdot y = x^T y = y^T x$

$\begin{pmatrix} 1 \\ 4 \\ 0 \\ 2 \end{pmatrix} \begin{pmatrix} 2 & -2 & 1 & 3 \end{pmatrix} = 2 - 8 + 0 + 6 = 0$

$\begin{pmatrix} 1 & 4 & 0 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ -2 \\ 1 \\ 3 \end{pmatrix} = 2 - 8 + 0 + 6 = 0$

$v_1^T v_2 = 0$ (orthogonal)

2. Show $v_1 = (1, 2, -1)$

$v_2 = (2, 1, 0)$

$\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} = 2 - 2 = 0$ [orthogonal]

2. Find k_3 that's orth to both $(1, 1, 0)$ & $(1, 0, 1)$

Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ find null space and special solⁿ is the ans.

$\downarrow R_2 - R_1$
 $\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

Ans: $\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$ & $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$

$x = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ $y = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$ $z = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$

4. Find vector x orth to row space(A),
 y orth to col space(A),
 z orth to null space

$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$

$x \perp C(A^T) \rightarrow x \in N(A)$
 $y \perp C(A) \rightarrow y \in N(A^T)$
 $z \perp N(A) \rightarrow z \in C(A^T)$

$R_2 - 2R_1 \downarrow R_3 - 3R_1$

$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} : b_1 \\ : b_2 - 2b_1 \\ : b_3 - 3b_1 \end{matrix}$

$\downarrow R_3 - R_2$

$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} : b_1 \\ : b_2 - 2b_1 \\ : b_3 - 3b_2 - b_1 \end{matrix}$

$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 2 & 4 & 6 \end{pmatrix} \rightarrow \begin{matrix} x \rightarrow \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \\ y \rightarrow \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \\ z \rightarrow \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \end{matrix}$

6. P in plane R^2 eq $\rightarrow x + 2y - z = 0$. Find vec $\perp$ to P .
 What matrix has P as its nullspace.
 What matrix has P as its row space.

$P = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix}, x + 2y - z = 0 \right\}$

$x + 2y - z = 0$

$\begin{pmatrix} 1 & 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$v^T w = 0$
 $v \perp w$

a) $(1, 2, -1) \perp P$

b) $\begin{pmatrix} 1 & 2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$A w = 0$

$A = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$
 matrix

$N(A) = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$

$N(A) = \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$

$P = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix}, x + 2y - z = 0 \right\}$

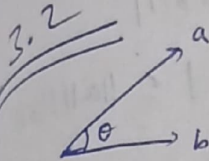
$x = -2y + z$

$P = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2y + z \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} y + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} z \text{ for all } y, z \in R \right\}$

$\Rightarrow N(A) = P$

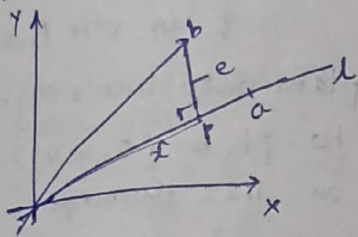
$$B = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$C(B^T) = P.$$



$$\cos \theta = \frac{a \cdot b}{\|a\| \|b\|} = \frac{a^T b}{\|a\| \|b\|}$$

Projection onto a line:-



'p' is a projection vector of 'b' on 'a'

$$p = \hat{a} a$$

, $\hat{a} = ?$

$$e = b - p$$

WKT, $a \perp e$

$$a^T e = 0$$

$$a^T (b - p) = 0$$

$$a^T b - a^T p = 0$$

$$a^T b - a^T \hat{a} a = 0$$

$$a^T b = a^T \hat{a} a$$

Scaling of projection

$$\hat{a} = \frac{a^T b}{a^T a}$$

dim $r \times c \rightarrow \text{scalar}$
 $c \times r \rightarrow \text{matrix}$

$a^T b$ - inner product of a, b

$a^T a$ - norm of a $\sim \|a\|^2$

$$\hat{a} = \frac{a(a^T b)}{a^T a}$$

$$= \left(\frac{a a^T}{a^T a} \right) b$$

$$p = P b$$

where $P = \frac{a a^T}{a^T a}$

is called projection of a matrix

Properties of P

rank 1

$$P^n = P$$

Symmetric

trace of P is one (Sum of diagonals) -

$a \rightarrow n$ -dim vec then P is square matrix of order n.

$a \rightarrow$ unit vector then $P = a a^T$.

* Schwarz inequality :-

$|a^T b| \leq \|a\| \|b\|$
all vectors a and b in $\mathbb{R}^n$ satisfy the Schwarz inequality.

→ equal when a and b are dependent.

$$\cos \theta = \frac{a^T b}{\|a\| \|b\|}$$

$$|\cos \theta| \leq 1$$

$$\left| \frac{a^T b}{\|a\| \|b\|} \right| \leq 1$$

$$|a^T b| \leq \|a\| \|b\|$$

3.2

Q. 1. Find the proj of $b = (1, 2, 7)$ onto $C(A)$ spanned by $(2, 1, -2)$ and $(1, -1, 4)$ split b into $p + q$ with p in $C(A)$ and q in Left Nullspace.

2. Matrix that projects every pt in plane onto line $x + 2y = 0$

3. Multiple of $a = (1, 1, 1)$ is closest to pt $b = (2, 4, 4)$?
Find also the pt closest to a on line through b .

4. What mat P projects every point in $\mathbb{R}^3$ onto the line of intersection of the planes $x + y + z = 0$ & $x - z = 0$?

② Given line, $x + 2y = 0$

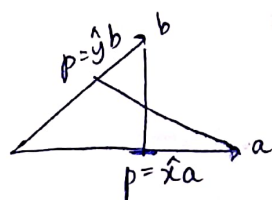
$$P = \frac{aa^T}{a^T a} \rightarrow \textcircled{1}$$

choose 'a' on $x + 2y = 0$

$$\text{Let } a = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$P = \frac{\begin{pmatrix} 2 \\ -1 \end{pmatrix} \begin{pmatrix} 2 & -1 \end{pmatrix}}{\begin{pmatrix} 2 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix}} = \frac{\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}}{5} = \begin{bmatrix} 4/5 & -2/5 \\ -2/5 & 1/5 \end{bmatrix}$$

③



$$\begin{aligned} a^T &= \text{row matrix} \\ a &= \text{col matrix} \\ \hat{x} &= \frac{a^T b}{a^T a} \end{aligned}$$

$$\frac{\begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix}}{\begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}} = \frac{10}{3}$$

$$a = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$b = \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix}$$

$$\hat{p} = \text{proj of } a \text{ on } b = \hat{y} b = \left(\frac{b^T a}{b^T b} \right) b$$

$$\begin{aligned} &= \frac{2}{18} \frac{\begin{pmatrix} 2 & 4 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}{\begin{pmatrix} 2 & 4 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix}} = \frac{10}{36} = \left(\frac{5}{18} \right) b \\ &= \frac{5}{18} \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 5/9 \\ 10/9 \\ 10/9 \end{bmatrix} \end{aligned}$$

Given, $x+y+t=0 \rightarrow P_1$
 $x-t=0 \rightarrow P_2$

Intersection of P_1, P_2 is a line.

$$\begin{array}{r} x+y+t \\ + \quad x \quad -t \\ \hline 2x+y=0 \end{array}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}; Ax=0$$

Sol is null space.

$x \in$ Nullspace.

As $x \in N(A)$,
 we have to find $N(A)$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \end{pmatrix} \xrightarrow{R_2 \div -1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$N(A) = \left\{ \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix} \right\}$$

$$R = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_1 - R_2} \begin{pmatrix} 1 & -1 & -3 \\ 0 & 1 & 2 \end{pmatrix}$$

The line of intersection of eq 1 & 2 is passing through the point $(1, -2, 1)$.

Let $a = (1, -2, 1)$

$$P = \frac{aa^T}{a^T a} = \frac{\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}_{3 \times 1} \begin{pmatrix} 1 & -2 & 1 \end{pmatrix}_{1 \times 3}}{\begin{pmatrix} 1 & -2 & 1 \end{pmatrix}_{1 \times 3} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}_{3 \times 1}} = \frac{\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}}{6}$$

$$P = \frac{1}{6} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

Trace is 1
 Symmetric
 Rank is 1.

$p = Ph$

Projections and least squares:

The failure of gaussian elim is alm certain when we have several eq in one unknown.

$$\begin{cases} a_1x = b_1 \\ \vdots \\ a_mx = b_m \end{cases} \text{ Solvable if } b = (b_1, \dots, b_m) \text{ multiple of } a = (a_1, \dots, a_m).$$

-Var

$$\begin{cases} 2x = 4 \\ 3x = 6 \\ 4x = 8 \end{cases} \text{ consi } \begin{cases} 2x = 4 \\ 3x = 7 \\ 4x = 9 \end{cases} \text{ inconsi}$$

$a = b$

$E = \|ax - b\|$

Parab
 min

$\hat{x}$ = estimate solution

"Error" $E^2 = (2x-4)^2 + (3x-7)^2 + (4x-9)^2$

$$E^2 = (a_1x - b_1)^2 + (a_2x - b_2)^2 + \dots + (a_mx - b_m)^2$$

$$\frac{d}{dx}(E^2) = 0$$

$$0 = 2(a_1x - b_1)a_1 + 2(a_2x - b_2)a_2 + \dots + 2(a_mx - b_m)a_m$$

$$0 = 2[a_1(a_1x - b_1) + a_2(a_2x - b_2) + \dots + a_m(a_mx - b_m)] = 0$$

$$0 = a_1^2x - b_1a_1 + a_2^2x - b_2a_2 + \dots + a_m^2x - b_ma_m = 0$$

$$(a_1^2 + a_2^2 + \dots + a_m^2)x = a_1b_1 + \dots + a_mb_m \Rightarrow x = \frac{a_1b_1 + \dots + a_mb_m}{a_1^2 + \dots + a_m^2}$$

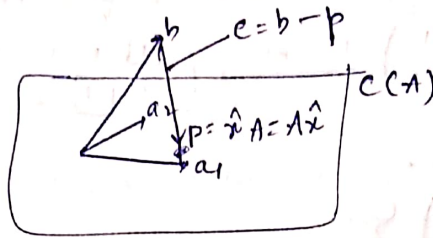
$$x = \frac{a^T b}{a^T a} = \hat{x} \text{ (is projection)} \quad b \text{ on } a$$

Sec var

$$Ax = b$$

$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{m \times n} x = b$$

$\hat{x}$ = least square solⁿ



$$e \perp C(A)$$

$$e \in N(A^T)$$

$$A^T e = 0$$

$$A^T(b - p) = 0$$

$$A^T b - A^T p = 0$$

$$A^T b - A^T A \hat{x} = 0$$

$$A^T b = A^T A \hat{x}$$

$$A^T A \hat{x} = A^T b$$

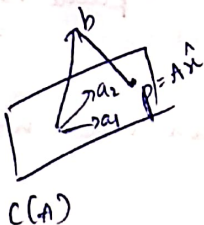
$$\hat{x} = (A^T A)^{-1} A^T b$$

normal eq

$$\hat{x} = \frac{A^T b}{A^T A}$$

X can't write why:
Bcoz $a^T a$ is scalar
 $A^T A$ is matrix.

Set
3.2
1.



$Ax = b$ is 'inconsistent' when $b \notin C(A)$

$$A \hat{x} = b$$

$$A^T A \hat{x} = A^T b \rightarrow \text{normal equation.}$$

$$\text{Here, } A = \begin{bmatrix} 1 & 1 \\ a_1 & a_1 \\ 1 & \phi \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -2 \\ 1 & 1 & 4 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}_{3 \times 2} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 1 & 1 & -2 \\ 1 & 1 & 4 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}_{3 \times 1}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & -2 \\ 1 & 1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -8 \\ -8 & 18 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} -11 \\ 27 \end{bmatrix}$$

$$6x_1 - 8x_2 = -11$$

$$-8x_1 + 18x_2 = 27$$

$$\hat{x}_1 = \frac{9}{22} \quad \hat{x}_2 = \frac{37}{22}$$

$$\begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 9/22 \\ 37/22 \end{bmatrix}$$

$$p = A \hat{x}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 9/22 \\ 37/22 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} \frac{9+37}{22} \\ \frac{9-37}{22} \\ \frac{-2(9)+4(37)}{22} \end{bmatrix} = \begin{bmatrix} 46/22 \\ -28/22 \\ 130/22 \end{bmatrix} = \begin{bmatrix} 23/11 \\ -14/11 \\ 65/11 \end{bmatrix}$$

$$q = b - p = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 23/11 \\ -14/11 \\ 65/11 \end{bmatrix} = \begin{bmatrix} -12/11 \\ 36/11 \\ 12/11 \end{bmatrix} = q$$

$$\boxed{\therefore b = p + q}$$

$$(C(A))^{\perp} \perp N(A)$$

Q. Find a basis for the orthogonal complement of the row space of A. Split $x = (3, 3, 3)$ into row space comp x_r & x_n

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix} = R$$

$$N(A) = \left\{ \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix} \right\}$$

$$x = x_r + x_n$$

Project $x = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$ on nullspace $\begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix} = b$ (say)

Project vector x on the null space.

$$\text{Proj of } x \text{ on } b = \left(\frac{b^T x}{b^T b} \right) b$$

$$x_n = \frac{(-2 \ -2 \ 1) \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}}{(-2 \ -2 \ 1) \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix}} \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix} = \frac{(-9)}{8} \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} = x_n$$

$$\text{Proj of } x \text{ on nullspace} = x_n = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$$x \text{ on row space} = x_r = x - x_n = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$$

$$x = x_n + x_r$$

$$\begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$$

Set
3.3

1. $Ax = b$ by least-squares, find $p = Ax$ if $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$
Verify that the error $b - p$ is $\perp$ to $C(A)$.

$$A^T A \hat{x} = A^T b$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}$$

$$p = A \hat{x} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ 2/3 \end{bmatrix}$$

$$b - p = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/3 \\ 1/3 \\ 2/3 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 2/3 \\ -2/3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow{R_3 - R_1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow C(A) = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$e \perp a_1 \therefore \begin{bmatrix} 2/3 \\ 2/3 \\ -2/3 \end{bmatrix} \perp \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad (e^T a_1 = 0)$$

$$\begin{pmatrix} 2/3 & 2/3 & -2/3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$e \perp a_2 \therefore \begin{bmatrix} 2/3 \\ 2/3 \\ -2/3 \end{bmatrix} \perp \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \quad (e^T a_2 = 0)$$

$$\begin{bmatrix} 2/3 & 2/3 & -2/3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = 0$$

Ex

3Q. Find proj of b onto $C(A)$. $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ -2 & 4 \end{pmatrix}$ $b = \begin{pmatrix} 1 \\ 2 \\ 7 \end{pmatrix}$

H.W
3/4

Split b into $p+q$ with p in col space & $q \perp$ to that space.
Which of the four subspaces contains q .

$$\rightarrow \text{Proj of } b \text{ on } a = \left(\frac{a^T b}{a^T a} \right) a = \left(\frac{a a^T}{a^T a} \right) b$$

Proj b on $C(A)$

$$p = A \hat{x}$$

$$= A(A^T A)^{-1} A^T b$$

$$p = P b$$

$$P = A(A^T A)^{-1} A^T \text{ projection matrix.}$$

5. Find proj matrix P onto the space spanned by $a_1 = (1, 0, 1)$ and $a_2 = (1, 1, -1)$

$$A = \begin{bmatrix} 1 & 1 \\ a_1 & a_2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}$$

$$P = A(A^T A)^{-1} A^T$$

$$A^T A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \Rightarrow (A^T A)^{-1} = \frac{1}{6} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix}$$

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & -1 \end{pmatrix}_{3 \times 2} \begin{pmatrix} 1/2 & 0 \\ 0 & 1/3 \end{pmatrix}_{2 \times 2} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix}_{2 \times 3} = \begin{pmatrix} 1/2 & 1/3 \\ 0 & 1/3 \\ 1/2 & -1/3 \end{pmatrix}_{3 \times 2} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix}_{2 \times 3}$$

$$\begin{bmatrix} 5/6 & 1/3 & 1/6 \\ 1/3 & 1/3 & -1/3 \\ 1/6 & 1/3 & 1/3 \end{bmatrix} = \begin{bmatrix} 1/2 + 1/3 & 1/3 & 1/2 - 1/3 \\ 1/3 & 1/3 & -1/3 \\ 1/6 & 1/3 & 1/3 \end{bmatrix}$$

Least Square Fitting of Data:-

$$(t_1, b_1), (t_2, b_2), \dots, (t_m, b_m) \quad y = a + bx$$

$$b = C + Dt$$

$$A\hat{x} = b$$

$$A^T A \hat{x} = A^T b$$

$$\hat{x} = \begin{bmatrix} C \\ D \end{bmatrix}$$

1. Use the method of least square to fit the best line to the data. $b = 4, 3, 1, 0$ at $t = -2, -1, 0, 2$. Find the projection of $b = (4, 3, 1, 0)$ onto the col space of A . Calculate the error vector 'e' and check that 'e' is orthogonal to the columns of A .

Let $C + Dt = b$ be the best fit straight line for the given data.

$$C + Dt = b$$

$$C + D(-2) = 4$$

$$C + D(-1) = 3$$

$$C + D(0) = 1$$

$$C + D(2) = 0$$

$$\begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix}_{4 \times 2} \begin{bmatrix} C \\ D \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 4 \\ 3 \\ 1 \\ 0 \end{bmatrix}_{4 \times 1}$$

$$Ax = b$$

$$A\hat{x} = b$$

$$A^T A \hat{x} = A^T b$$

$$\hat{x} = \begin{pmatrix} 4 & -1 \\ -1 & 9 \end{pmatrix}^{-1} \begin{pmatrix} 8 \\ -11 \end{pmatrix}$$

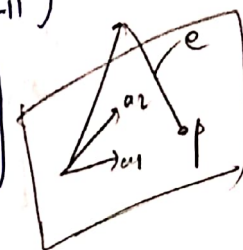
$$= \frac{1}{35} \begin{pmatrix} 9 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 8 \\ -11 \end{pmatrix}$$

$$\hat{x} = \frac{1}{35} \begin{pmatrix} 61 \\ -36 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 2 \end{pmatrix}_{2 \times 4} \begin{pmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{pmatrix}_{4 \times 2} \begin{pmatrix} C \\ D \end{pmatrix}_{2 \times 1} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 2 \end{pmatrix}_{2 \times 4} \begin{pmatrix} 4 \\ 3 \\ 1 \\ 0 \end{pmatrix}_{4 \times 1}$$

$$\begin{pmatrix} 4 & -1 \\ -1 & 9 \end{pmatrix} \begin{pmatrix} C \\ D \end{pmatrix} = \begin{pmatrix} 8 \\ -11 \end{pmatrix}$$

$$\hat{x} = \begin{pmatrix} 61/35 \\ -36/35 \end{pmatrix}$$



Best fit line $b = \frac{61}{35} - \frac{36}{35}t$

$$p = A\hat{x} = \frac{1}{35} \begin{pmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 61 \\ -36 \end{pmatrix} = \frac{1}{35} \begin{pmatrix} 133 \\ 97 \\ 61 \\ -11 \end{pmatrix} = p \in \mathbb{R}^4$$

$$\rightarrow e = b - p = \begin{pmatrix} 4 \\ 3 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{35} \begin{pmatrix} 133 \\ 97 \\ 61 \\ -11 \end{pmatrix} \Rightarrow e = \begin{pmatrix} 1/5 \\ 8/35 \\ -26/35 \\ 11/35 \end{pmatrix} \quad e \in N(A^T)$$

$$\rightarrow a_1 \perp e \quad (\because a_1^T e = 0) \quad (1, 1, 1, 1) \cdot e = 0 = \frac{1}{5} + \frac{8}{35} - \frac{26}{35} + \frac{11}{35}$$

$$a_2 \perp e \quad (\because a_2^T e = 0) \quad (-2, -1, 0, 2) \cdot e = 0 = -\frac{2}{5} - \frac{8}{35} + \frac{22}{35}$$

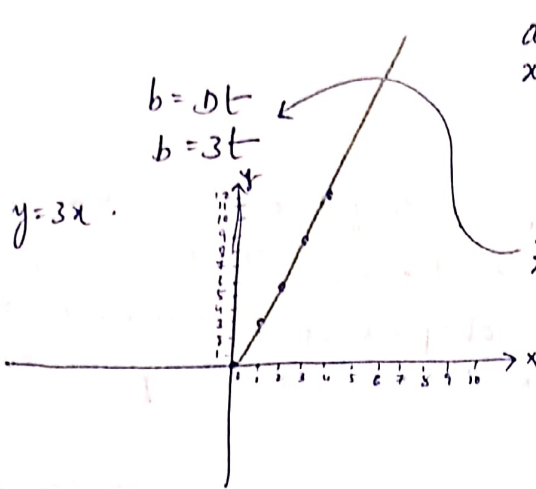
2Q. $b_1 = 1$ $t_1 = 1$ $b = Dt$ } origin passing
 $b_2 = 7$ $t_2 = 2$ solve $D = 1$,
 $2D = 7$ by least squares.

$$b = Dt$$

$$1 = D(1)$$

$$7 = D(2)$$

$$\begin{bmatrix} 1 & 1 \\ 7 & 2 \end{bmatrix} [D] = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$



$$ax = b$$

$$x^1 = \text{proj of } b \text{ on } a$$

$$= \frac{a^T b}{a^T a} = \frac{(1 \ 2) \begin{pmatrix} 1 \\ 7 \end{pmatrix}}{(1 \ 2) \begin{pmatrix} 1 \\ 2 \end{pmatrix}} = \frac{15}{5} = 3$$

$$x = 3 = D$$

* Weighted Least Squares :-

Scale 1 $x = 45$
 Scale 2 $x = 48$

$$\begin{matrix} x = b_1 \\ x = b_2 \end{matrix} \} \begin{bmatrix} 1 \\ 1 \end{bmatrix} x = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Single var :- $ax = b$
 $(wa)x = (wb)$

$$\hat{x} = \frac{(wa)^T wb}{(wa)^T (wa)}$$

$$\hat{x} = \frac{(1 \ 1) \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}}{(1 \ 1) \begin{pmatrix} 1 \\ 1 \end{pmatrix}} = \frac{b_1 + b_2}{2}$$

$$\star \boxed{\hat{x} = \frac{b_1 + b_2}{2}}$$

Mul var :- W

$$(WA)x = (Wb)$$

$$(WA)^T (WA) \hat{x} = (WA)^T (Wb)$$

$$(A^T W^T) (WA) \hat{x} = (A^T W^T) (Wb)$$

$W = \text{diagonal matrix.}$

$$\boxed{A^T (W^T W) A \hat{x} = A^T (W^T W) b} \text{ Weighted normal equation}$$

Q. Find the weighted least squares solⁿ x^1 wth $Ax = b$.

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \quad b = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \quad W = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$W^T W = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A^T W^T W b = \begin{pmatrix} 4 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$A^T W^T W = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$A^T W^T W A = \begin{pmatrix} 4 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 3 & 5 \end{pmatrix}$$

$$A^T W^T W A \hat{x} = A^T W^T W b$$

$$\begin{pmatrix} 6 & 3 \\ 3 & 5 \end{pmatrix} \hat{x} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$= \frac{1}{21} \begin{pmatrix} 5 & -3 \\ -3 & 6 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \frac{1}{21} \begin{pmatrix} 1 \\ 12 \end{pmatrix} \Rightarrow \hat{x} = \begin{pmatrix} 1/21 \\ 12/21 \end{pmatrix}$$

eq. $\hat{x}w = \frac{1}{21} \begin{pmatrix} 1 \\ 12 \end{pmatrix}$



* $\begin{matrix} 3, 4 \\ (a, b, c) \end{matrix} \xrightarrow{\text{col of } A} \begin{matrix} \text{col of } Q \\ (q_1, q_2, q_3) \end{matrix} ?$

independent orthonormal vectors $\rightarrow$ every vector has length 1 & is orthonormal to each other in a set.

$\rightarrow \underline{E}_x :=$ col of Identity matrix, $\mathbb{R}^2$.

$\rightarrow$ preserve length, orientation, angle.

$Q = \begin{bmatrix} | & | & | \\ q_1 & q_2 & \dots \\ | & | & | \end{bmatrix}$ square, rect, also

• has col as orthonormal vectors.

• $Q^T Q = I$

• $Q^{-1} = Q^T$

• preserve angle if norm

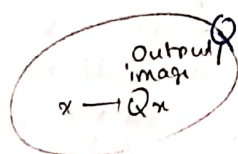
$Ax = b \Rightarrow x = A^{-1}b$

$Qx = b \Rightarrow x = Q^{-1}b$

$x = Q^T b$

easier.

$\rightarrow x \in \mathbb{R}^n \quad \|x\|^2 = x^T x$



$Q^T Q = I$

$Q^T Q^T Q = Q^T I Q^{-1}$

$Q^T (Q^T Q) = Q^{-1}$

$Q^{-1} = Q^T$

$x, y \in \mathbb{R}^n$

$\langle Qx, Qy \rangle = (Qx)^T Qy$

$= x^T Q^T Qy$

$= x^T y$

$\langle x, y \rangle = x^T y = x \cdot y$

$\therefore$ inner prod preserved.

$\|Qx\|^2 = (Qx)^T (Qx)$

$= x^T Q^T Qx$

$= x^T I x$

$= x^T x$

$\|Qx\|^2 = \|x\|^2 \therefore$ preserves norm.

$\rightarrow \cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|} \Rightarrow \cos \phi = \frac{\langle Qx, Qy \rangle}{\|Qx\| \|Qy\|} \Rightarrow \boxed{\theta = \phi}$

* $x \in \mathbb{R}^n$

consider an orthonormal basis $\{q_1, q_2, \dots, q_n\}$

$x = x_1 q_1 + x_2 q_2 + \dots + x_n q_n \xrightarrow{\text{①}}$

q_i - col m
 q_i^T - row m

Multiply ① by q_i^T on b.s

$q_i^T x = x_1 (q_i^T q_1) + x_2 (q_i^T q_2) + \dots + x_n (q_i^T q_n)$

$= x_1 + 0 \dots$

$q_1^T x = x_1$

$q_2^T x = x_2$

$x_n = q_n^T x$

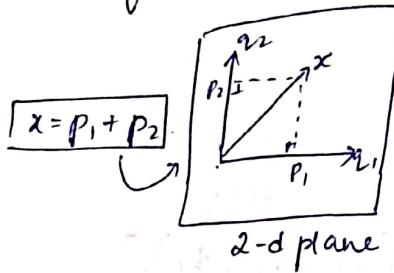
Sub in ①

$$x = (q_1^T x) q_1 + q_2^T q_2 x + \dots + q_n^T x q_n$$

$$= \text{proj of } x \text{ on } q_1 + (\text{proj of } x \text{ on } q_2) + \dots + (\text{proj of } x \text{ on } q_n)$$

$$= \text{Sum of projections of } x \text{ on } q_i$$

$$x = p_1 + p_2 + \dots + p_n$$



Proj of b on a

$$= \left(\frac{a^T b}{a^T a} \right) a$$

$$= \frac{a^T b}{a^T a} a$$

x on q1

$$= \left(\frac{q_1^T x}{q_1^T q_1} \right) q_1$$

$$= (q_1^T x) q_1$$

$$Ax = b$$

(follow) b comb of col of A.

b can be exp as sum of proj

(not follow)

$$\hat{b} = A \hat{x}$$

$$\hat{b} = \hat{p}_1 + \hat{p}_2 + \hat{p}_3 + \dots + \hat{p}_n$$

(p1) proj of b on q1 +

(p2) " " q2 +

(p3) " " qn

$$A^T A \hat{x} = A^T b$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$\hat{b} = A \hat{x}$$

$$\hat{b} = A (A^T A)^{-1} A^T b$$

lengthy.

$$A \rightarrow Q ? \text{ (How) }$$

Q := Check whether rotational matrix is orthogonal

Permutation matrix is orthogonal.

projection matrix not orthogonal.

$$\text{Rot} = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}$$

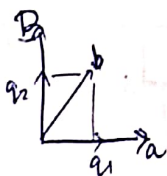
$$Q^T Q = \begin{pmatrix} c & -s \\ s & c \end{pmatrix} \begin{pmatrix} c & s \\ -s & c \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \text{orthogonal}$$

$$H^T H = I = \text{orthogonal}$$

* Gram-Schmidt Process:-

a set of independent vectors $\rightarrow$ a set of orthonormal vectors

$$a, b, c \rightarrow q_1, q_2, q_3$$



$$q_1 = \frac{a}{\|a\|}$$

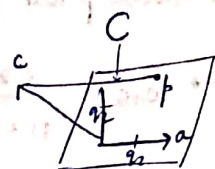
$$B = b - \text{proj of } b \text{ on } q_1$$

$$b - (q_1^T b) q_1$$

$$q_2 = \frac{B}{\|B\|}$$

To find q_3 :-

$$C' = c - \text{proj of } c \text{ on col space of } q_1, q_2$$



$$C' = c - (p_1 + p_2)$$

$$= c - [(q_1^T c) q_1 + (q_2^T c) q_2]$$

$$= c - q_1^T q_1 c - q_2^T q_2 c$$

$$q_3 = \frac{C'}{\|C'\|}$$

$$C' = c - q_1^T c q_1 - q_2^T c q_2$$

have to normalise

$D = d - \text{proj of } d \text{ on col space of } a_1, a_2, a_3$
 $D = d - (p_1 + p_2 + p_3)$
 $= d - a_1^T d a_1 - a_2^T d a_2 - a_3^T d a_3$
 $q_4 = \frac{D}{\|D\|}$

4.11. From the non-orthogonal a, b, c , find the q_1, q_2, q_3
 $a = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad c = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$q_1 = \frac{a}{\|a\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \cancel{q_2 = \frac{b}{\|b\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}} \quad \cancel{q_3 = \frac{c}{\|c\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}}$

$q_2 = \frac{B}{\|B\|} \Rightarrow B = b - (q_1^T b) q_1$
 $= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$
 $= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{bmatrix} 0 \\ 1/2 \\ 1/2 \end{bmatrix} = B = \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}$

$\|B\| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + 1} = \sqrt{\frac{2}{4} + 1} = \frac{\sqrt{6}}{\sqrt{2}} = \frac{\sqrt{3}}{\sqrt{2}} = \sqrt{\frac{3}{2}}$

$q_2 = \frac{B}{\|B\|} = \sqrt{\frac{2}{3}} \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{\sqrt{3} \cdot 2} \\ \frac{-\sqrt{2}}{\sqrt{3} \cdot 2} \\ \frac{\sqrt{2}}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{\sqrt{2}}{\sqrt{3}} \end{pmatrix} = q_2$

$q_3 = \frac{C}{\|C\|} \Rightarrow C = c - q_1^T c q_1 - q_2^T c q_2$

$= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix} - \frac{1}{\sqrt{6}} \begin{pmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{pmatrix}$

$= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1/2 \\ 1/2 \\ 0 \end{pmatrix} - \begin{pmatrix} 1/6 \\ -1/6 \\ 2/6 \end{pmatrix}$

$= \begin{pmatrix} -1/3 \\ 1/3 \\ 1/3 \end{pmatrix}$

$\|C\| = \frac{2}{\sqrt{3}}$

$\begin{pmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$

$q_1^T c = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}$
 $q_2^T c = \begin{pmatrix} 1/\sqrt{6} & -1/\sqrt{6} & \sqrt{2}/\sqrt{3} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}} + \frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}}$

$\begin{pmatrix} 1/\sqrt{6} & -1/\sqrt{6} & \sqrt{2}/\sqrt{3} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}} + \frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}}$

$\begin{pmatrix} 1/\sqrt{6} & -1/\sqrt{6} & \sqrt{2}/\sqrt{3} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}} + \frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}}$

$\begin{pmatrix} 1/\sqrt{6} & -1/\sqrt{6} & \sqrt{2}/\sqrt{3} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}} + \frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}}$

$\sqrt{\left(-\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2}$

$= \sqrt{\frac{1}{9} + \frac{1}{9} + \frac{1}{9}} = \frac{\sqrt{3}}{3}$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 1 & 1 \\ q_1 & q_2 & q_3 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 1/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}$$

$$A = QR$$

$$R = Q^T A = Q^T A$$

$$= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{6} & -1/\sqrt{6} & 1/\sqrt{6} \\ -1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 3/\sqrt{6} & 1/\sqrt{6} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix} = \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ 0 & q_2^T b & q_2^T c \\ 0 & 0 & q_3^T c \end{bmatrix}$$

$$\begin{aligned} &\rightarrow Ax = b \\ &A = LU \\ &A = QR \end{aligned}$$

$$QRx = b$$

$$Rx = Q^T b$$

$$\boxed{Rx = Q^T b}$$

R less comp than A.
time

- ① Find an orthonormal set q_1, q_2, q_3 for which q_1, q_2 span the col space of $A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$. which find subspace contains q_3 ? What is the least squares sol of $Ax = b$ if $b = [1 \ 2 \ 7]^T$.

$$A = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

$$q_1 = \frac{a}{\|a\|} = \frac{1}{3} \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 2/3 \\ -2/3 \end{bmatrix}$$

$$\|a\| = \sqrt{1^2 + 2^2 + (-2)^2} = \sqrt{1+4+4} = 3$$

$$q_2 = \frac{B}{\|B\|}$$

$$B = b - (q_1^T b)q_1$$

$$= \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 2 \end{bmatrix}$$

$$q_2 = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$$

$$q_1^T b = \begin{bmatrix} 1/3 & 2/3 & -2/3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

$$= \frac{1}{3} - \frac{2}{3} - \frac{8}{3} = -3$$

$$= -3 \begin{bmatrix} 1/3 \\ 2/3 \\ -2/3 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 2 \end{bmatrix}$$

$$\|B\| = \sqrt{2^2 + 2^2 + 1^2} = 3$$

$$q_3 = \begin{pmatrix} y \\ z \end{pmatrix}$$

$$x^2 + y^2 + z^2 = 1$$

$$q_1 \perp q_3 \Rightarrow q_1^T q_3 = 0$$

$$q_2 \perp q_3 \Rightarrow q_2^T q_3 = 0$$

$$q_3 = C - (q_1^T C) q_1 - q_2^T C q_2$$

$$\begin{pmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

$$A q_3 = 0$$

$$q_3 \in N(A)$$

Gauss multiplication method.

$$\begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{pmatrix}$$

$$q_1^T q_3 = 0$$

$$\begin{pmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$x + 2y - 2z = 0$$

$$q_2^T q_3 = 0$$

$$\begin{pmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$2x + y + 2z = 0$$

Eigen vector & Eigen values :-

$$Ax = \lambda x$$

λ Eigen value of corresponding — scaling factors of vectors
 x Eigen vector of A

$$A_{n \times n} x_{n \times 1} = \lambda x$$

transformed ones are Eigen vector

$$Ax - \lambda x = 0$$

$$x(A - \lambda I) = 0 \quad \text{System of linear equations.}$$

$$(A - \lambda I)x = 0 \quad \text{Homogeneous.}$$

Solve it $\rightarrow$ Eigen vectors.

$$\text{In form of } Bx = 0$$

Have solution $x = 0$ (trivial solution)

$B \neq 0$ (must be singular)
 $B \uparrow$ matrix

$$|B| = 0$$

$$|A - \lambda I| = 0$$

$\lambda = \lambda_1, \lambda_2, \lambda_3, \dots$ we'll get $\lambda^n + ()\lambda^{n-1} + \dots = 0$

Characteristic eqⁿ
 Eigen eqⁿ

Solving $\rightarrow$ Eigen value/vector.

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 0 - \lambda & 1 \\ -2 & -3 - \lambda \end{vmatrix} = 0$$

$$= \begin{vmatrix} -\lambda & 1 \\ -2 & -3 - \lambda \end{vmatrix} = -\lambda(-3 - \lambda) - (-2)(1) = 0$$

$$+3\lambda + \lambda^2 + 2 = 0$$

$$\lambda^2 + 3\lambda + 2 = 0$$

$$\lambda = -1, \lambda = -2$$

Case 1, $\lambda_1 = -2$

$$(A - \lambda I)x = 0$$

$$(A + 2I)x = 0$$

$$\left(\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \right) x = 0$$

$$\begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\begin{aligned} 2x_1 + x_2 &= 0 \\ -2x_1 - x_2 &= 0 \end{aligned}$$

Null space of $A - \lambda I$ matrix.

$$\begin{cases} 2x_1 + x_2 = 0 \\ -2x_1 - x_2 = 0 \end{cases}$$

$R_1 + R_2$

$$\begin{pmatrix} 1 & 1/2 \\ 0 & 0 \end{pmatrix} = R_1$$

$$x = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}$$

$$x_1 = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \quad \text{just to avoid decimal. } k=2$$

Case 2, $\lambda_2 = -1$

$$(A - \lambda I)x = 0$$

$$(A + I)x = 0$$

$$\left[\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] x = 0 \Rightarrow \begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\begin{aligned} x_1 + x_2 &= 0 \\ -2x_1 - 2x_2 &= 0 \end{aligned}$$

$$\begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix} \xrightarrow{R_1 + R_2} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = R_2 \quad x_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

zero matrix entire $\mathbb{R}^2$ Eigen vector

dif eigen vectors have dif eigen values,

Ex: $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

$$|A - \lambda I| = 0$$

$$\left| \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

Case 1: $\lambda_1 = 1$

$$(A - \lambda I)x = 0$$

$$(A - I)x = 0$$

$$\left[\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] x = 0$$

$$\begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \rightarrow \begin{pmatrix} 2 & 4 \\ 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} = R$$

Case 2: $\lambda_2 = 5$

$$(A - \lambda I)x = 0$$

$$\left[\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} \right] x = 0$$

$$\begin{pmatrix} -4 & 4 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \rightarrow \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} = R$$

$$x_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$\begin{aligned} &\begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} \\ &\downarrow R_2 + \frac{1}{2}R_1 \\ &\begin{bmatrix} -4 & 4 \\ 0 & 0 \end{bmatrix} \cdot \frac{1}{-4} \\ &\begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$\begin{cases} 2x_1 + x_2 = 0 \\ -2x_1 - x_2 = 0 \end{cases}$$

$$A + R_2$$

$$\begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix} = R_2$$

$$X_1 = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}$$

Just to avoid decimal.

$$X_1 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$k=2$

$$\begin{aligned} 1) x_1 &= 0 \\ -2) x_2 &= 0 \\ +x_2 &= 0 \\ -2x_2 &= 0 \end{aligned}$$

$$e_2 \quad X_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$(A - \lambda I)x = 0$$

$$Ax = 0$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$0 \cdot x_1 + 0 \cdot x_2 = 0$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)^2 = 0$$

$$4 + \lambda^2 - 4\lambda = 0$$

$$\lambda = 2, 2$$

Case 1: $\lambda_1 = 2$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} 2-2 & 0 \\ 0 & 2-2 \end{pmatrix} x = 0$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

2 free variables.

Quant. $A + I = \begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix}$

$$(A - \lambda I)x = 0$$

$$\begin{vmatrix} 2-\lambda & 4 \\ 2 & 4-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2-\lambda & 4 \\ 2 & 4-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(4-\lambda) - 8 = 0$$

$$8 - 2\lambda - 4\lambda + \lambda^2 - 8 = 0$$

$$\lambda^2 - 6\lambda = 0$$

$$\lambda = 6, 0$$

Case 1: $\lambda_1 = 6$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} x = 0$$

$$\begin{pmatrix} 2-\lambda & 4 \\ 2 & 4-\lambda \end{pmatrix} x = 0$$

$$\begin{pmatrix} -4 & 4 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \rightarrow \begin{pmatrix} -4 & 4 \\ 2 & -2 \end{pmatrix} \xrightarrow{R_2 + \frac{1}{2}R_1} \begin{pmatrix} -4 & 4 \\ 0 & 0 \end{pmatrix}$$

2 free variables.

20. $A = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

$$B = \begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Check.

$$\lambda_1 + \lambda_2 + \lambda_3 = \text{trace}$$

$$\lambda_1 \lambda_2 \lambda_3 = \text{determinant}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 3-\lambda & 4 & 2 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & -\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3-\lambda & 4 & 2 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & -\lambda \end{vmatrix} = 0$$

$$3 - \lambda[(1-\lambda)(-\lambda)] - 4(0) + 2(0)$$

$$(3-\lambda)[- \lambda + \lambda^2]$$

$$-3\lambda + 3\lambda^2 + \lambda^2 - \lambda^3$$

$$\lambda = 0, 1, 3$$

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad |A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)^2 = 0$$

$$\lambda = 2, 2$$

Case 1: $\lambda_1 = 2$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} 2-2 & 1 \\ 0 & 2-2 \end{pmatrix} x = 0$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

Case 1: $\lambda = 0$

$$(A - \lambda I)x = 0$$

$$(A - 0)x = 0$$

$$\begin{pmatrix} 3 & 4 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\downarrow R_1 - 4R_2$$

$$\begin{pmatrix} 3 & 0 & -6 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -4x_2 \\ x_2 \\ 0 \end{pmatrix}$$

$$3x + 4y + 2z = 0$$

$$0x + 1y + 2z = 0$$

Using cross multiplⁿ,

$$\frac{x}{4} = \frac{y}{-6} = \frac{z}{3}$$

$$\begin{vmatrix} 4 & 2 \\ 1 & 2 \end{vmatrix} \quad \begin{vmatrix} 2 & 3 \\ 2 & 0 \end{vmatrix} \quad \begin{vmatrix} 3 & 4 \\ 0 & 1 \end{vmatrix}$$

$$\frac{x}{6} = \frac{y}{-6} = \frac{z}{3}$$

$$\Rightarrow \frac{x}{2} = \frac{y}{-2} = \frac{z}{1}$$

$$X_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}^k, k \in \mathbb{R}.$$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

$$(A - \lambda I)x = 0$$

$$(A - I)x = 0$$

$$\begin{pmatrix} 2 & 4 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\begin{cases} 2x + 4y + 2z = 0 \\ 0x + 0y + 2z = 0 \\ 0x + 0y - z = 0 \end{cases}$$

$$\begin{cases} 2x + 4y + 2z = 0 \\ 0x + 0y + 2z = 0 \\ 0x + 0y - z = 0 \end{cases}$$

Using cross multiplⁿ,

$$\frac{x}{4} = \frac{y}{2} = \frac{z}{0}$$

$$\frac{x}{8} = \frac{y}{-4} = \frac{z}{0}$$

$$\Rightarrow \frac{x}{2} = \frac{y}{-1} = \frac{z}{0}$$

$$X_2 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$$

Case 3: $\lambda = 3$

$$(A - \lambda I)x = 0$$

$$(A - 3I)x = 0$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 0 & -2 & 2 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\begin{cases} 0x + 4y + 2z = 0 \\ 0x - 2y + 2z = 0 \\ 0x + 0y - 3z = 0 \end{cases}$$

$$\begin{cases} 0x + 4y + 2z = 0 \\ 0x - 2y + 2z = 0 \\ 0x + 0y - 3z = 0 \end{cases}$$

Using cross multiplⁿ,

$$\frac{x}{4} = \frac{y}{2} = \frac{z}{0}$$

$$\frac{x}{12} = \frac{y}{0} = \frac{z}{0}$$

$$X_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -\lambda & 0 & 2 \\ 0 & 2-\lambda & 0 \\ 2 & 0 & -\lambda \end{vmatrix} = 0$$

$$-\lambda(-\lambda + \lambda^2) - 0 + 2(0 - 2(2-\lambda))$$

$$+\lambda^3 + 2(-4 + 2\lambda)$$

$$-\lambda^3 + \lambda^2 - 8 + 4\lambda = 0$$

$$4\lambda^2(2-\lambda) - 4(2-\lambda) = 0$$

$$2-\lambda[\lambda^2 - 4] = 0$$

$$\lambda^2 = 4$$

$$\lambda = \pm 2$$

$$\lambda = 2, 2, -2$$

Case 1: $\lambda_1 = 2$

$$(B - \lambda I)x = 0$$

$$(B - 2I)x = 0$$

$$\begin{pmatrix} -2 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$-2x + 0y + 2z = 0$$

$$2x + 0y - 2z = 0$$

$\begin{matrix} x \\ y \\ z \end{matrix} \neq \begin{matrix} 0 \\ 0 \\ 0 \end{matrix}$ } only when independent rows are there.

$\rightarrow x = z, y \text{ is arbitrary.}$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad x_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

infinitely many
max 2 indep eigen vectors.

Case 2: $\lambda_2 = -2$

$$(B + 2I)x = 0$$

$$\begin{pmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

when eigen value repeat

$$\begin{cases} 0x + 4y + 0z = 0 \\ 2x + 0y + 2z = 0 \end{cases}$$

$$\frac{x}{0} = \frac{y}{2} = \frac{z}{2}$$

$$\frac{x}{+8} = \frac{y}{0} = \frac{z}{-8}$$

$$x_3 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 2 & 2 \\ 2 & 2-\lambda & 2 \\ 2 & 2 & 2-\lambda \end{vmatrix} = 0$$

$$2-\lambda[(2-\lambda)^2 - 4] - 2[2(2-\lambda) - 4] + 2[4 - 2(2-\lambda)] = 0$$

3x3 matrix
std characteristic matrix.

$$\lambda^3 - (\sum d)\lambda^2 + (\sum M_d)\lambda - |A| = 0$$

$$\sum d = 2 + 2 + 2 = 6$$

$$\sum M_d = \begin{vmatrix} 2 & 2 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 2 & 2 \end{vmatrix} = 0$$

$$|C| = 0$$

$$\lambda^3 - 6\lambda^2 = 0$$

$$\lambda = 0, 6$$

Case 1: $\lambda_1 = 0$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$x_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Case 2: $\lambda_2 = 6$

$$(A - 6I)x = 0$$

$$\begin{pmatrix} 1-6 & 2 & 2 \\ 2 & 2-6 & 2 \\ 2 & 2 & 2-6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\begin{pmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$-4x + 2y + 2z = 0$$

$$2x - 4y + 2z = 0$$

$$\frac{x}{2} = \frac{y}{-4} = \frac{z}{2}$$

$$\frac{x}{12} = \frac{y}{12} = \frac{z}{12}$$

$$x_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 3 & 0 & -\lambda \end{vmatrix} = 0$$

$$-\lambda(-2\lambda + \lambda^2) - 0 + 1(0 - 6 + 3\lambda)$$

$$+ 2\lambda^2 - \lambda^3 - 6 + 3\lambda$$

$$\lambda^3 - (2\lambda^2) + (3\lambda - 6) = 0$$

$$\lambda^2(\lambda - 2) + 3(\lambda - 2)$$

$$\lambda^2(\lambda - 2) + 3(\lambda - 2)$$

$$\lambda = -\sqrt{3}, 2, \sqrt{3}$$

Case 1 $\lambda = -\sqrt{3}$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} +\sqrt{3} & 0 & 1 \\ 0 & 2+\sqrt{3} & 0 \\ 3 & 0 & +\sqrt{3} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\frac{x}{2+\sqrt{3}} = \frac{y}{0} = \frac{z}{3}$$

Prop:

$$\det(A - \lambda I) = 0$$

$$\det(A) = \lambda_1 \lambda_2 \dots$$

$$\frac{x}{(2+\sqrt{3})(\sqrt{3})} = \frac{y}{0} = \frac{z}{+3(2+\sqrt{3})}$$

$$\frac{x}{+2\sqrt{3}+3} = \frac{y}{0} = \frac{z}{+6+3\sqrt{3}}$$

Case 2: $\lambda = 2$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} -2 & 0 & 1 \\ 0 & 0 & 0 \\ 3 & 0 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\frac{x}{-2} = \frac{y}{0} = \frac{z}{3}$$

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{0}$$

$$x_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

Case 3: $\lambda = \sqrt{3}$

$$(A - \lambda I)x = 0$$

$$\begin{pmatrix} -\sqrt{3} & 0 & 1 \\ 0 & 2-\sqrt{3} & 0 \\ 3 & 0 & -\sqrt{3} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$\frac{x}{-2\sqrt{3}+3} = \frac{y}{0} = \frac{z}{+6+3\sqrt{3}}$$

Properties:

$$S = \begin{bmatrix} x_1 & x_2 & x_3 \\ | & | & | \end{bmatrix}$$

$$AS = SA$$

$$\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix}$$

Post mul S^{-1}

$$A S S^{-1} = S \Lambda S^{-1}$$

$$A = S \Lambda$$

Pre mul S^{-1}

$$S^{-1} A S = \Lambda$$

$$A^2 = A \cdot A$$

$$= S \Lambda (S^{-1} S) \Lambda S^{-1}$$

$$= S \Lambda \mathbb{I} \Lambda S^{-1}$$

$$A^2 = S \Lambda^2 S^{-1}$$

$$A^3 = S \Lambda^3 S^{-1}$$

$$A^k = S \Lambda^k S^{-1}$$

$$f(\lambda) = 0$$

$$f(A) = 0$$

$$\lambda^2 - 3\lambda - 18 = 0$$

$$A^2 - 3A - 18\mathbb{I} = 0$$

$$A^{-1} A^2 - 3A^{-1} A - 18A^{-1} \mathbb{I} = A^{-1} 0$$

$$A - 3\mathbb{I} - 18A^{-1} = 0$$

$$18A^{-1} = A - 3\mathbb{I}$$

$$A^{-1} = \frac{1}{18} [A - 3\mathbb{I}]$$

Unit Test

Cayley Hamilton
Theorem
(C-H) Theorem

$$\lambda = x$$